

Reduced-order modeling and calculation of vortex-induced vibration for large-span bridges



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ABSTRACT

The Volterra theory-based reduced-order modeling (ROM) technique is utilized for simulating fluid-structure interactions of bridge decks during vortex-induced vibration (VIV). To improve the order of nonlinearity adopted in the Volterra series, sparse form kernels are adopted in this study. To illustrate and validate the performance of ROM in simulating fluid-structure interactions of VIV, a real bridge deck is used as a case study. The kernels in the Volterra series are identified using least squares with input–output pairs obtained from a CFD approach. Results indicate that ROM can well reproduce the inherent nature of nonlinearities existing in fluid-structure interactions described by the CFD approach. A method for calculating the VIV of bridges by combining ROM with the structural finite element model is also proposed. The VIV performance of a real bridge is obtained through the proposed method and is compared with field measurements.

1. Introduction

Large-span bridges are quite sensitive to wind effects. Since the collapse of the Tacoma Narrows Bridge, there has been great development in the subject of bridge wind resistance. However, there are still some aspects that need to be further improved, among which is the prediction of vortex-induced vibration (VIV) for a real bridge at the concept design stage.

The most direct way to achieve this object is to conduct wind tunnel experiments on an aeroelastic model that has the same dynamics and configurations as a real bridge, or to use the CFD approach with three-dimensional (3D) grids and arbitrarily shaped moving boundaries. The involving of a real 3D bridge in the CFD approach will be difficult to implement in the foreseeable future. In addition, experimental results of an aeroelastic model can seldom be comparable with those of field measurements or sectional models, since the scale ratio of an aeroelastic model is usually much smaller than that of a sectional model. Thus, wind tunnel experiments or CFD calculations of VIV are normally conducted on a sectional model with structural parameters determined according to the equivalent modal mass of the deck for a certain mode.

However, the VIV response of a structure is sensitive to a structural parameter: the mass-damping parameter. For large-span bridges, there are numerous VIV modes in the velocity range of interest. The modal mass and modal damping vary for each mode, which would require a

mountain of experimental work or CFD calculations on sectional models in order to obtain the full VIV performance of a bridge. Furthermore, a real bridge is a flexible structure with 3D modal shapes whose VIV response varies with that of the sectional model. Moreover, oncoming flows are not always uniformly distributed along the span-wise axis of the bridge. In addition, 3D modal shapes may cause vortices to move along the span-wise axis. All these 3D effects of flow fields may cause a loss of coherence of VIV forces along the span-wise direction, and result in inconsistency between the VIV responses of the sectional model and the real bridge.

A promising way to overcome these problems is to utilize mathematical models to model the flow-structure interactions existing in VIV and to combine these models with the structural finite element model to calculate the VIV responses of a bridge under different modal conditions.

Since the work of Bishop and Hassan (1964), a lot of these kinds of models have been proposed, which can be roughly divided into two main categories: single-degree-of-freedom models and two-degree-of-freedom models. Single-degree-of-freedom models can be further classified into negative-damping models (D'Asdia et al., 2003; Ehsan and Scanlan, 1990; Goswami et al., 1993; Larsen, 1995; Marra et al., 2011; Scanlan, 1998) and force-decomposition models (Griffin, 1980; Iwan and Botelho, 1985; Sarpkaya, 1978). Two-degree-of-freedom models can also be further divided into two main subclasses: wake-oscillator models (Facchinetti et al., 2004; Farshidianfar and

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Zanganeh, 2010; Skop and Balasubramanian, 1997) and Milan-oscillator models (Diana et al., 2006; Falco et al., 1999).

The goal of a mathematical model for VIV should be prediction for a certain deck shape for values of mass-damping parameters different from that for which the model parameters were estimated. Unfortunately, none of the above empirical models seem to be able to accomplish this task, since they always have arbitrarily assumed forms that hardly touch the intrinsic nature of fluid-structure interactions.

Recently, some conceptually novel and computationally efficient techniques have been proposed for computing unsteady flows and fluid-induced effects, e.g. reduced order modeling (ROM) techniques. Among the various ROMs, the Volterra theory-based model provides the possibility to catch the inherent nature of nonlinearities and memory effects existing in fluid-structure interactions. The basic premise of this model is that a large class of nonlinear systems can be modeled as a sum of multidimensional convolution integrals of increasing order (Volterra et al., 1959).

The Volterra theory was first proposed and successfully applied in electrical engineering, and was then utilized to model non-Gaussian fluid loadings and mechanical nonlinearities as well as fluid-structure interactions in the fields of offshore engineering (Rugh, 1981; Schetzen, 2006). After that, Silva (1997) applied it in modeling the aeroelastic phenomenon in aerospace engineering. Recently Wu and Kareem introduced it into bridge wind engineering to model aerodynamics and aeroelasticity (Wu and Kareem, 2013a, 2015). Wu and Kareem (2013b) studied the prospect of Volterra series in modeling the VIV system of a bridge. However, the VIV system in their study is represented by the traditional nonlinear semi-empirical model, which is still a far cry from practical application.

In this paper, the Volterra theory-based reduced-order model is applied in practice by involving a real bridge. A procedure for calculating the VIV response of a bridge by combining ROM with the structural finite element model is also proposed. This paper is organized as follows: Section 2 briefly introduces the reduced order modeling of VIV and its kernel identification method; Section 3 applies ROM to a real bridge deck to model the fluid-structure interactions existing in VIV; Section 4 proposes a procedure for calculating the VIV of bridges by combining ROM with the structural finite element method, and the VIV performance of a real bridge is obtained by using this procedure; and Section 5 gives some concluding remarks.

2. Reduced-order modeling and kernel identification

2.1. Reduced-order modeling of bridge deck VIV

For bridge decks submerged in flow fields, as discussed by Wu and Kareem (2013a), motion-induced effects under a stationary uniform wind flow (constant wind velocity) can be uniformly expressed by taking the vertical motion velocity $\dot{y}$ and the rotary motion θ and velocity $\dot{\theta}$ as the input information of bridge aerodynamics:

$$F(t) = f(\theta, \dot{y}, \dot{\theta}, t) \quad (1)$$

where, f denotes a generalized function. The VIV of a bridge deck often occurs in a single degree of freedom. Taking the vertical degree of freedom as an illustration, the lift force on a strip of deck can be expressed as:

$$F(t) = \frac{1}{2} \rho U^2 D C_L(\dot{y}, t) \quad (2)$$

where, $C_L(\dot{y}, t)$ denotes normalized lift forces; D is deck height; U is the oncoming wind velocity; and y is the structural vertical displacement from the static deformation position.

The VIV system can typically be treated as time-invariant, causal and with fading memory. For such a system, the lift force can be uniformly represented as:

$$C_L(t) = \int_0^t h_1(\tau) \dot{y}(t-\tau) d\tau + \int_0^t \int_0^t h_2(\tau_1, \tau_2) \dot{y}(t-\tau_1) \dot{y}(t-\tau_2) d\tau_1 d\tau_2 + \sum_{p=3}^M \int_0^t \dots \int_0^t h_p(\tau_1, \dots, \tau_p) \dot{y}(t-\tau_1) \dots \dot{y}(t-\tau_p) d\tau_1 \dots d\tau_p \quad (3)$$

where, $h_1(\tau)$ is the first-order kernel, which describes the linear behavior of the system; $h_p(\tau_1, \dots, \tau_p)$ is the p th-order kernel term, which describes the p th-order nonlinearity existing in the system; and M is the degree of nonlinearity to be considered in the VIV system.

In real practice, the time series of both $\dot{y}(t)$ and $C_L(t)$ generated either from experiments or CFD calculations are discrete time series. Thus, Eq. (3) is always written in discrete-time form, which can be expressed as:

$$C_L[k] = \sum_{p=1}^M H_p[k] \\ H_p[k] = \sum_{m_1=0}^{M_p} \dots \sum_{m_p=0}^{M_p} h_p[m_1, \dots, m_p] \prod_{j=1}^p \dot{y}[k-m_j] \quad (4)$$

Eq. (4) can be rewritten in matrix form:

$$\mathbf{C}_{L \times 1} = \mathbf{Y}'_{N \times Q} \mathbf{H}_{Q \times 1} \quad (5)$$

where $N = n + 1$ denotes the length of the input-output pairs; Q denotes the length of the kernel vector $\mathbf{H}_{Q \times 1}$ in the discrete-time model; $\mathbf{C}_{L \times 1}$ is a vector of the output data; and $\mathbf{Y}'_{N \times Q}$ is a matrix which consists of the input data combinations.

$$\mathbf{C}_{L \times 1} = [C_L[0], C_L[1], \dots, C_L[n]]^T$$

$$\mathbf{Y}'_{N \times Q} = \begin{bmatrix} Y'^1_{N \times Q_1}, Y'^2_{N \times Q_2}, \dots, Y'^p_{N \times Q_p}, \dots, Y'^M_{N \times Q_M} \end{bmatrix}$$

$$\mathbf{H}_{Q \times 1} = \begin{bmatrix} \mathbf{H}^1_{Q_1}, \dots, \mathbf{H}^p_{Q_p}, \dots, \mathbf{H}^M_{Q_M} \end{bmatrix}^T \quad (6)$$

Since the summation notation for the Volterra operators in Eq. (6) can be quite complicated, for illustration purposes, an explicit expression for a particular Volterra series expansion is provided in the Appendix A.

2.2. Kernel identification

The application of the Volterra series in engineering fields can be roughly classified into two main categories. The first is to represent a dynamic phenomenon that has complicated analytical expressions, so the kernels can be generated by using the harmonic probing method. The other is to build a model for an observed dynamic phenomenon that does not have any analytical expression, so the kernels are estimated from experimentally or numerically generated data.

The reduced order modeling of VIV using the Volterra series belongs to the latter case, so experimentally or numerically generated data is needed. Traditionally in aerodynamic and aeroelastic applications, kernel identification has been performed using impulsive inputs. However, the identification of Volterra kernels through impulse-input approach is a resource-intensive endeavor. For kernels that are larger than 3, the analytical details for input-output relations become increasingly fussy, and the interpolation idea involving impulses of various weights becomes increasingly barren from a feasibility viewpoint. Thus, the identification of kernels through impulse-input approach is always limited to the second-order. In fact, recent studies indicate that the impulse identification method is far from optimal in terms of accuracy and efficiency comparing with the least-square approach, especially when higher-order kernels are involved in this process (Balajewicz and Dowell, 2012; Balajewicz et al., 2012).

It is well known that the even-order terms capture the even-order super-harmonics and thus take into account the asymmetric nonlinearities, whereas the odd-order terms capture the odd-order super-harmonics and thus take into account the symmetric nonlinearities.

The bridge deck is always asymmetric with respect to the chord line, so the lowest-order nonlinear model for simulating the bridge aerodynamics should have a degree of three (consisting of the first, second and third order kernels). Since the bridge exhibits limit cycle oscillation during VIV, the order of nonlinearity to be considered for comprehensively simulating the fluid-structure interactions during VIV should be higher than 3. Thus, the least-square procedure is adopted in this study for kernel identification, since it is easier to identify higher-order kernels comparing with the impulse-input approach.

According to Eq. (5), the kernel term $\mathbf{H}_{Q \times 1}$ can be rewritten as:

$$\mathbf{H}_{Q \times 1} = (\mathbf{Y}'_{N \times Q})^+ \mathbf{CL}_{N \times 1} \quad (7)$$

where symbol “+” denotes the Moore-Penrose pseudo-inverse. If the input and output series are longer than the kernel term ($N > Q$), Eq. (7) is a procedure of least-square fit.

2.3. Sparse form of Volterra model

In general, the performance of the identification is proportional to the length of the training input or the sufficiency of information it contains. The input-output pair in Eq. (7) should be an assembly of data generated under different states of motions of the bridge deck, for example, at different reduced amplitudes and frequencies. The kernel term generated by Eq. (7) can then be used to model the fluid-structure interactions within these states.

However, as indicated in Eq. (6) and the Appendix A, the kernel terms increase geometrically with the degree of nonlinearity of the Volterra model, so the computational cost associated with kernel identification also grows geometrically. This rapid growth of complexity and computational cost limits most applications of the Volterra model up to the second order.

Fortunately, in recent years, numerous methods have been proposed to alleviate the computational cost. Most of these methods are pruning schemes utilized for simplifying the Volterra model, for example, eliminating the insignificant kernels, or adopting the triangular kernel form or a widely used scheme where only the diagonal terms of the standard Volterra kernel are used (Wu and Kareem, 2014).

For aeroelasticity of bridge decks, experiments indicate that the frequency multiplications existing in the wind force as an output of a harmonic deck movement or gust wind are integral multiples of the input frequency (Diana et al., 2006; Zhu et al., 2013). Thus, the pruned form of the full-order Volterra kernel with only diagonal terms can still be considered to be in accord with the inherent nature of nonlinearity existing in bridge aerodynamics and aeroelasticity. The computational cost to identify higher-order sparse Volterra kernels is much lower than is required to identify lower-order full Volterra kernels. Since higher-order kernels are associated with stronger nonlinearities, identification of higher-order sparse kernels can often be more beneficial than identification of lower-order full kernels.

Similar to Eq. (4), the sparse Volterra model can be expressed as:

$$\begin{aligned} C_L[k] &= \sum_{p=1}^M D_p[k] \\ D_p[k] &= \sum_{m_p=0}^{M_p} h_p[m_p, \dots, m_p] \dot{y}[k - m_p]^p \end{aligned} \quad (8)$$

The matrix form of Eq. (8) can be expressed as:

$$\mathbf{CL}_{N \times 1} = \mathbf{Y}'_{N \times Q} \mathbf{D}_{Q \times 1} \quad (9)$$

where the input data matrix $\mathbf{Y}'_{N \times Q}$ for the sparse form Volterra model and the kernel vector $\mathbf{D}_{Q \times 1}$ are provided in explicit expression in the Appendix A.

The kernels of the sparse form Volterra model are not generated by firstly identifying the kernels of the full Volterra model and retaining only the diagonal terms. Instead, they are identified by directly using

the pseudo-inverse least-square fitting method described in Section 2.2:

$$\mathbf{D}_{Q \times 1} = (\mathbf{Y}'_{N \times Q})^+ \mathbf{CL}_{N \times 1} \quad (10)$$

The sparse kernels generated by Eq. (10) can compensate for the absence of off-diagonal terms due to pruning on the full Volterra model in a least-square fitting sense.

3. Modeling VIV of Xihoumen Bridge

In order to illustrate and validate the Volterra modeling technique in simulating the VIV of bridge decks, the real bridge deck of the Xihoumen Bridge was used in this study. This bridge is composed of two separated steel box girders spaced 6 m apart. It is located on the East China Sea, and is a suspension bridge with a main span of 1650 m and two side spans of 578 m.

The kernels in the Volterra model for this deck shape are identified by using the least-square-fit approach with input-output pairs generated from experiments or CFD calculations. In this paper, for the sake of demonstrating the effectiveness of the Volterra-based approach in modeling the VIV of bridge decks, the kernels are identified with data generated by a two-dimensional (2D) CFD platform, originally developed at the SLDRCE of Tongji University. Details of the 2D CFD platform and its careful validations can be found in Ge and Liu (2014) and Liu (2014).

The computational domain size is chosen as: $-15.5B \leq L_x \leq 34.5B$ and $-15B \leq L_y \leq 15B$. The computational domain is discretized with an unstructured mesh (Fig. 1). The distance from the first grid layer to the structural surface is $0.001B$, and the total number of cells of the computational domain is $1e5$. This choice of mesh was based on several mesh refinement studies carried out by Liu (2014). The Reynolds number in this study is set to $2.7e4$ with respect to deck height (3.5 m). The large eddy simulation (LES) with a Smagorinsky subgrid scale model is employed in the computational process.

For kernel identification, the input is the time-dependent vertical boundary velocity of the moving rigid body and the primary output is selected to be the integrated lift force on the structure. The input-output pairs in different forced-vibration states are calculated by the 2D CFD platform. The total number of forced-vibration cases is 165, with 15 reduced wind velocities (in the range of $0.428 \leq U/\omega_n D \leq 1.372$) and 11 reduced amplitudes (in the range of $0.01 \leq A/D \leq 0.11$). For each forced-vibration case, the input-output pair has a total time of 50 vibration periods. The input-output pairs for different forced-vibration cases can then be assembled in Eq. (7) as follows:

$$\mathbf{H}_{Q \times 1} = \begin{bmatrix} \mathbf{Y}'_{N_1 \times Q} \\ \vdots \\ \mathbf{Y}'_{N_k \times Q} \\ \vdots \\ \mathbf{Y}'_{N_{165} \times Q} \end{bmatrix}^+ \begin{bmatrix} \mathbf{CL}_{N_1 \times 1} \\ \vdots \\ \mathbf{CL}_{N_k \times 1} \\ \vdots \\ \mathbf{CL}_{N_{165} \times 1} \end{bmatrix} \quad (11)$$

where $\mathbf{Y}'_{N_k \times Q}$ and $\mathbf{CL}_{N_k \times 1}$ are the input data matrix and the output vector obtained from the k th forced-vibration case, respectively.

3.1. Determination of parameters in the Volterra model

To conduct the identification procedure, two parameters in the Volterra model should be determined in advance: the degree of

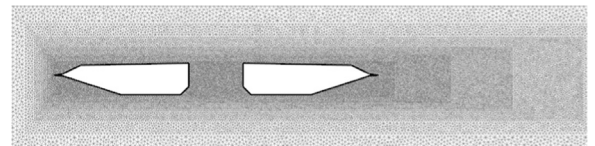


Fig. 1. Mesh near deck surface in computational domain.

nonlinearity M and the duration M_p for the p th-order term.

The duration M_p denotes the duration of the fluid memory effects of the aerodynamic forces. Wu and Kareem (2014) studied the duration of fluid memory effects on streamlined deck sections described by the Küssner function and suggested that for bridge decks the duration of the linear fluid memory effects can be set to 20 non-dimensional time, and the duration of the $(p + 1)$ th-order nonlinear fluid memory effects ($p \geq 1$) can be set to half the duration of the p th-order term.

As to the degree of nonlinearity in the Volterra model, as stated above, the order of nonlinearity to be considered for comprehensively simulating the fluid-structure interactions during VIV should be higher than 3. Theoretically speaking, the higher the degree of nonlinearity to be considered, the better the modeling performance will be. However, as is discussed in Section 2.3, the involving of higher-order terms will result in higher computational cost. Thus, the determination of the degree of nonlinearity should be a balance of accuracy and efficiency.

To detect the performance of the Volterra model in simulating the fluid-structure interactions during VIV, the concept of percent relative error, as defined by Balajewicz and Dowell (2012), is adopted in this study:

$$\%error = \frac{\|y_{CFD} - y_{ROM}\|_2}{\|y_{CFD}\|_2} \cdot 100 = \left(\frac{\sum \|y_{CFD}[n] - y_{ROM}[n]\|^2}{\sum \|y_{CFD}[n]\|^2} \right)^{1/2} \cdot 100 \quad (12)$$

The percent relative error of the full Volterra model versus the degree of nonlinearity M at different duration times are illustrated in Fig. 2. The non-dimensional time interval in the calculation is $0.1s$ ($s = Ut/D$). Thus, the non-dimensional duration time for the p th-order kernel is $\tau_p = 0.1M_p$. As illustrated by this figure, the percent relative error decreases with the increase of both the duration time, M_p , and the degree of nonlinearity, M . However, once the duration time is larger than $M_{p(p=1,2,\dots)}=50$, the increase of M will lead to a better modeling performance than the increase of M_p . Since the computational cost in the identification of the full Volterra kernels grows geometrically with the increase of the degree of nonlinearity, the identification is limited to the 3rd-order in this case.

The percent relative error of the Sparse form Volterra model versus the degree of nonlinearity M at different duration times are illustrated in Fig. 3. Similar with the full Volterra model, the percent relative error also decreases with the increase of both the duration time and the degree of nonlinearity. Besides, once the duration time is larger than $M_{p(p=1,2,\dots)}=50$, the increase of M will also lead to a better modeling performance than the increase of M_p . Since the computational cost in the identification of the Sparse form Volterra kernels grows linearly with the increase of the degree of nonlinearity, higher-order kernels are identified in this case.

The results indicate that once the duration time is large enough ($\tau_p = 0.1 \times 50 = 5$, in this case), the increase of degree of nonlinearity will lead to a better modeling performance than the increase of M_p . In addition, a higher-order Sparse-form Volterra model has a better modeling performance than a lower-order full Volterra model.

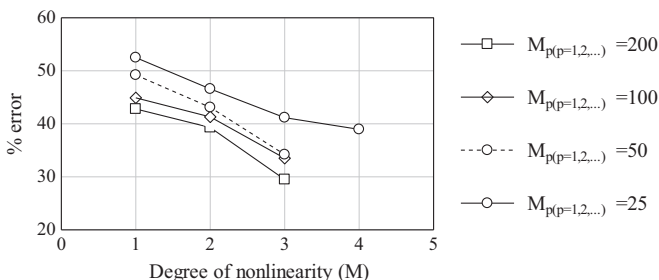


Fig. 2. Percent relative error of full Volterra kernel model versus M .

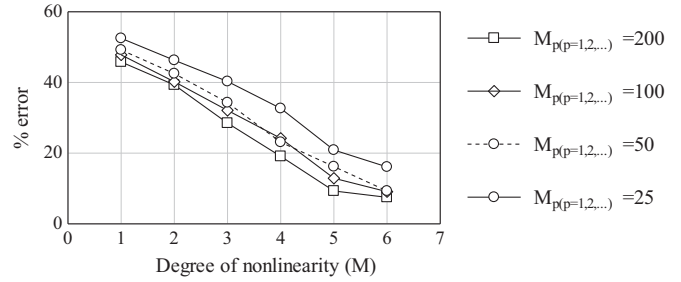


Fig. 3. Percent relative error of sparse Volterra kernel model versus M .

3.2. Kernel identification and validation

The results in the above section indicate that the modeling performance of a higher-order Sparse Volterra model is better than that of a lower-order full Volterra model. Thus, in this study, the Sparse-form Volterra model is adopted for modeling the fluid-structure interactions during VIV. The first- and second-order terms τ_1 and τ_2 are set to 20 and 10 non-dimensional times, respectively, and for the other higher-order terms, the non-dimensional duration time τ_p ($p \geq 3$) are set to 5 to sufficiently take into account the higher order nonlinear effects existing in the VIV of bridge decks.

The percent relative error versus the degree of nonlinearity in this case is illustrated by Fig. 4. As is indicated, the modeling performance improves with the increase of the degree of nonlinearity. However, the rate of improvement decreases at the same time. Once the degree of nonlinearity is larger than 6 or 7, the rate of improvement becomes very small. Thus, the degree of nonlinearity, M , in this study is set to 7.

The identified first- and second-order kernels are shown in Fig. 5. It is noticeable that both of these kernels vanish versus non-dimensional time, which indicates that the fluid memory effects fade away quickly after a period of time.

It should be noticed that the solution of Eq. (10) may face the problem of ill-condition, where the output of Eq. (10), $\mathbf{D}_{Q \times 1}$, are sensitive to changes or errors in the input, $\mathbf{C}_{L \times N \times 1}$. The measured inputs-outputs of the VIV system are inevitably influenced by noise due to equipment or environment in real practice. Thus, if the kernel identification of Eq. (10) is faced with the ill-conditioned problem, the identified results will be far from the actual results.

The ill-conditioned problem of a matrix can be roughly assessed by using the concept of condition number. Consider the over-determined system:

$$Ax = b \quad (13)$$

where the matrix $A \in \mathbb{R}^{m \times n}$. The traditional condition number in the 2-norm is defined as:

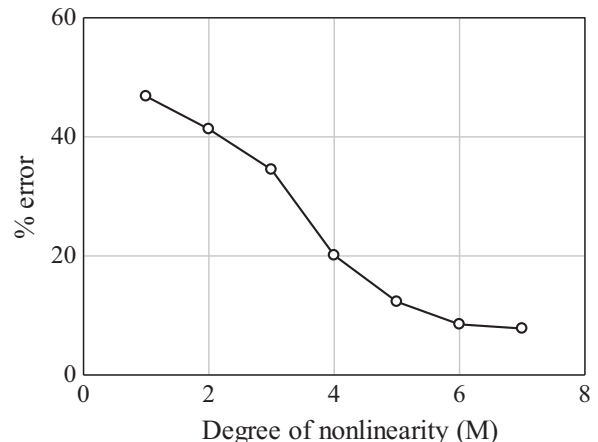
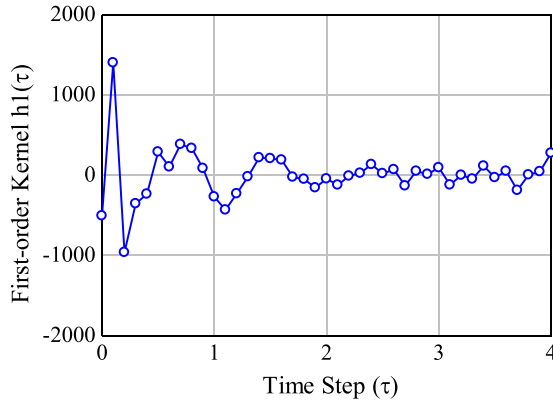
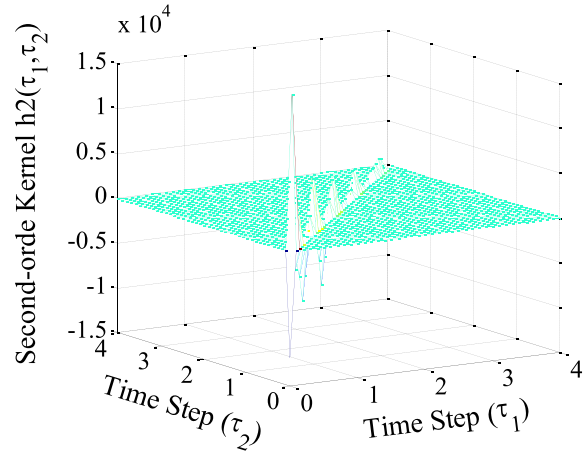


Fig. 4. Percent relative error versus M .



(a) first-order kernel



(b) second-order kernel

Fig. 5. Identified first- and second-order kernel. (a) first-order kernel (b) second-order kernel.

$$\kappa(A) = \frac{\sigma_{\max}}{\sigma_{\min}} \quad (14)$$

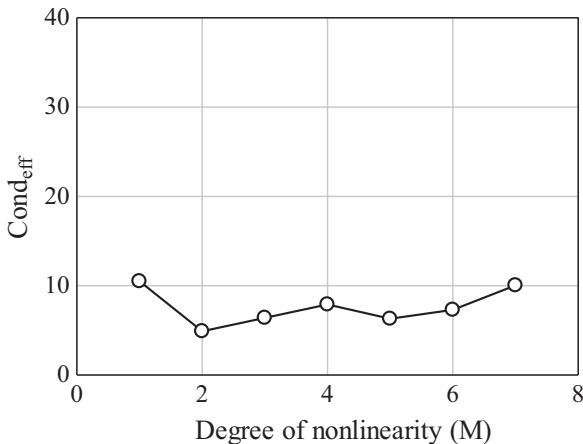
where $\sigma_{\max}$ and $\sigma_{\min}$ are the maximal and the minimal singular values of A , respectively.

A system with a low condition number is said to be well-conditioned, while a system with a high condition number is said to be ill-conditioned. And a system is said to be singular if the condition number is infinite. The condition number serves as a rough estimate of how many digits are lost in solving a linear system with that matrix. As a rule of thumb, if the condition number $\kappa(A) = 10^k$, then you may lose up to k digits of accuracy on top of what would be lost to the numerical method due to loss of precision from arithmetic methods (Cheney and Kincaid, 1999).

However, recent studies indicate that this simple normwise measure may overestimate by several orders of magnitude the true numerical condition of the least-square (LS) problem. Thus, the effective condition number is proposed for better stability analysis of LS problem (Li et al., 2010; Winkler, 2007):

$$\text{Cond}_{\text{eff}} = \frac{\|b\|}{\sigma_{\min} \|x\|} \quad (15)$$

The effective condition number of Eq. (10) versus degree of nonlinearity, M , is illustrated by Fig. 6. The calculated maximum effective condition number of the LS system is around 10, which is much smaller than the magnitude of the kernel terms. This indicates that the LS problem of Eq. (10) is well conditioned, and the errors due

Fig. 6. Effective condition number versus M .

to LS algorithm itself can be neglected in this case.

The identified kernels are then utilized to simulate the VIV forces on the bridge deck under different vibration states and compared with the original CFD results at the same conditions. Fig. 7 illustrates the time histories of non-dimensional lift force simulated by the Volterra model and the original CFD results. The nonlinear characteristics of the lift force vary with wind velocity and structural amplitude. And the phase lag between the high-frequency and the fundamental frequency components also changes with the increase of vibration amplitude at a certain wind velocity (see $A/D=0.05$ and $A/D=0.09$). The change of this phase lag with the increase of structural amplitude may be the reason for the occurrence of LCO phenomenon. The time histories simulated by the Volterra model compare well with those of the original CFD approach, which indicates that the Sparse Volterra model has well captured the dominant inherent nature of nonlinearities existing in the fluid-structure interactions described by the CFD approach.

Fig. 8 illustrates the hysteresis loops of the lift force versus structural velocity. The hysteresis loop is a useful tool to judge the phase lag between the aeroelastic force and the structural vibration status. The results indicate that the simulated hysteresis loops by the Volterra model also compare well with those of the CFD approach.

Fig. 9 illustrates the spectrum of the non-dimensional lift force simulated by the Volterra model and the original CFD approach. The results of these two approaches also compare well with each other. The frequency multiplications in the lift force are integer multiples of the structural vibration frequency, which validates that the use of pruned form of full-order Volterra kernel with only diagonal terms (Sparse-form Volterra model) are in accord with the inherent nature of nonlinearity existing in the bridge aeroelasticity.

3.3. Calculating VIV at different mass-damping parameters

The modeling performance of the Volterra model is validated through comparison between the simulated results and original CFD approach. In this section, the kernels identified through data obtained from forced-vibration conditions are utilized to calculate the free-vibration responses of the deck and compared with the CFD free-vibration results.

The dimensionless equation of motion for the deck in the vertical degree of freedom can be written as:

$$\begin{bmatrix} Y(s) \\ Y'(s) \end{bmatrix} = \begin{bmatrix} Y'(s) \\ -2\xi_n K_n Y'(s) - K_n^2 Y(s) + \frac{\rho D^2}{2M} C_L(s) \end{bmatrix} \quad (16)$$

where $Y = y/D$ is reduced displacement; ξ_n is mechanical damping; M is

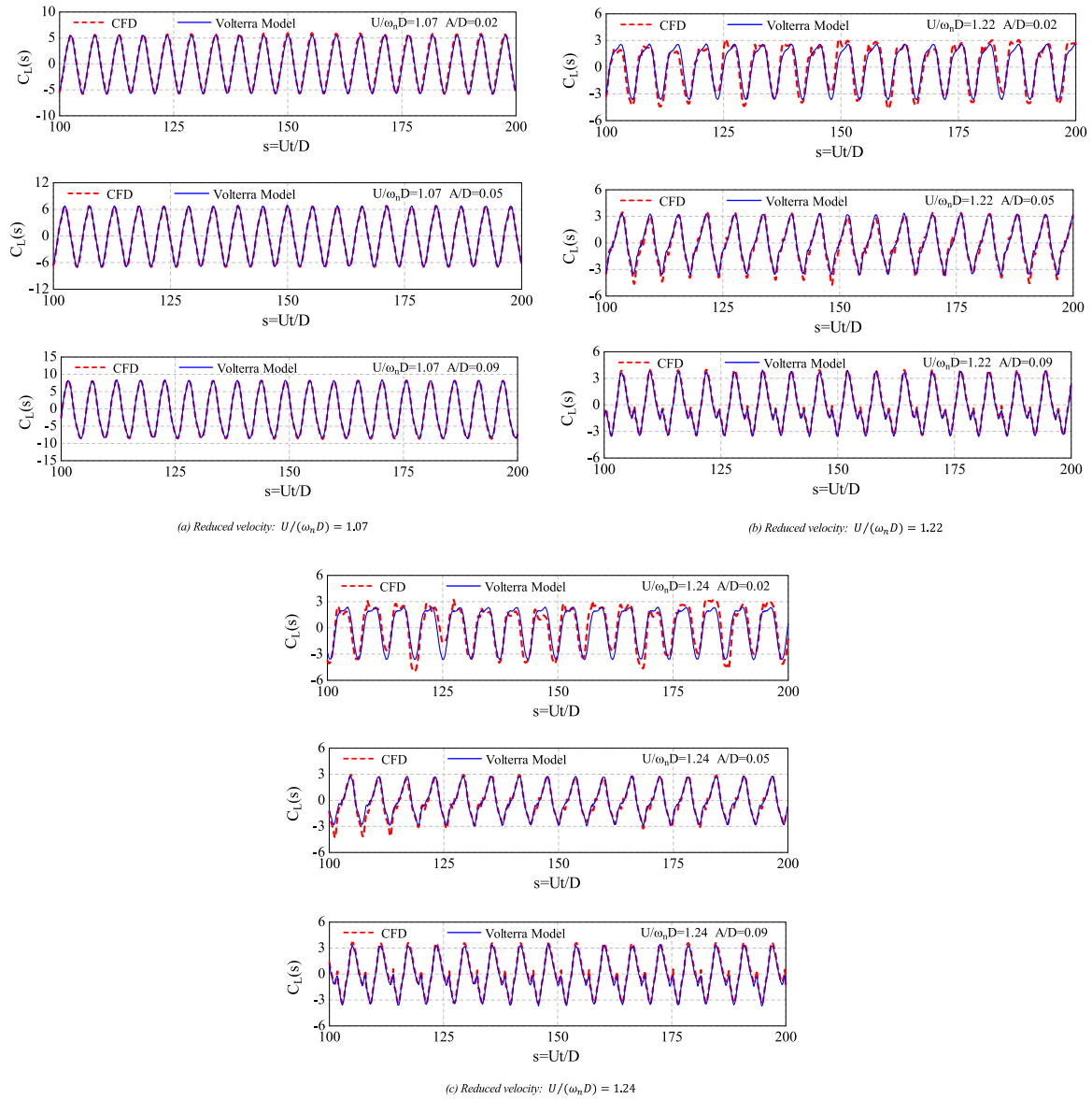


Fig. 7. Comparison between sparse Volterra modeling results and CFD results. (a) Reduced velocity: $U/(\omega_n D) = 1.07$. (b) Reduced velocity: $U/(\omega_n D) = 1.22$. (c) Reduced velocity: $U/(\omega_n D) = 1.24$.

structural mass; $s = Ut/D$ is non-dimensional time; $K_n = \omega_n D/U$ is reduced natural frequency; and $C_L(s)$ is the lift force coefficient described by the sparse Volterra model as defined by Eq. (8).

Eq. (16) can be solved by the 4th-order Runge Kutta method, where the dimensionless time interval is 0.1. The initial condition in the calculation is set to $[Y(0), Y'(0)] = [0, 0.005]^T$. $C_L(0)$ is first generated through Eq. (8) with the initial condition and is then applied to Eq. (16) to calculate the first step of structural motion: $[Y(1), Y'(1)]^T$. The first step of lift force, $C_L(1)$, can then be obtained through Eq. (8) with the calculated first step of structural motion, and is then adopted in calculating the next step of structural motion. By iterating this process, the structural VIV responses can be generated step by step.

To validate the calculated results of the Volterra model, four mass-damping cases are considered in the calculation, as illustrated in Table 1.

Taking Case D as an illustration ($M = 26393 \text{ kg}$, $\xi_n = 0.5\%$), The calculated VIV responses under two wind velocities in and out of the lock-in region are illustrated in Fig. 10. The results indicate that the Volterra model with kernels identified through forced-vibration data can well simulate the free-vibration features of the VIV system in and

out of the lock-in region (LCO in the lock-in region and decaying-vibration out of the lock-in region).

The calculated free-vibration displacements by the Volterra model are compared with those of the CFD approach at the same mass-damping conditions. Fig. 11 illustrates the results generated by these two approaches. As is indicated by Fig. 11, the calculated results by the Volterra model out of the lock-in region is decaying-vibration, and the results by CFD approach is small-amplitude signature-turbulence-induced vibration, the amplitude of which can be neglected comparing with the amplitude of VIV. While, in the lock-in region, these two sets of displacements both gradually grow to a steady-vibration state from a small amplitude initial condition. Besides, the reduced vibration frequencies of these two sets of displacements are almost identical and are slightly different from the structural natural frequency. This phenomenon of frequency shifting is also widely observed in wind tunnel experiments.

However, in the “transient” stage, at which the VIV gradually develops from the initial condition to the steady state, these two sets of displacement are not consistent with each other. The length of the “transient” stage of the CFD approach is shorter than that of the

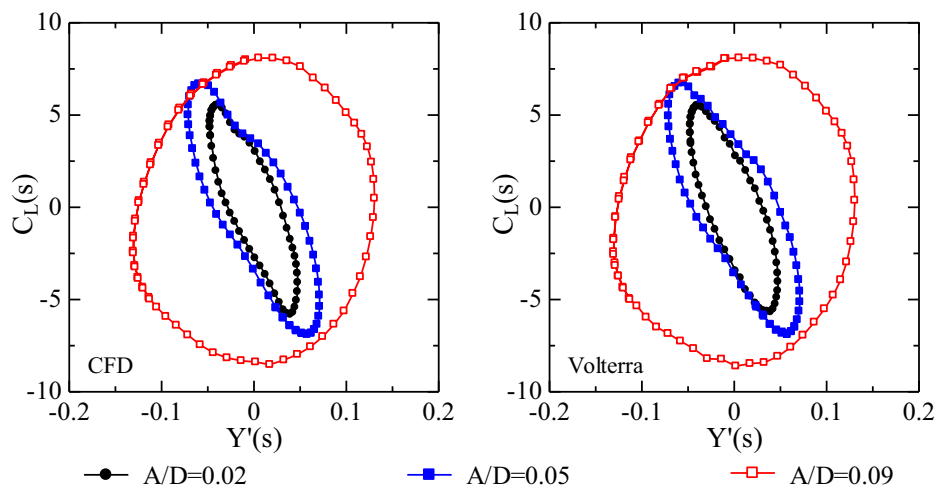
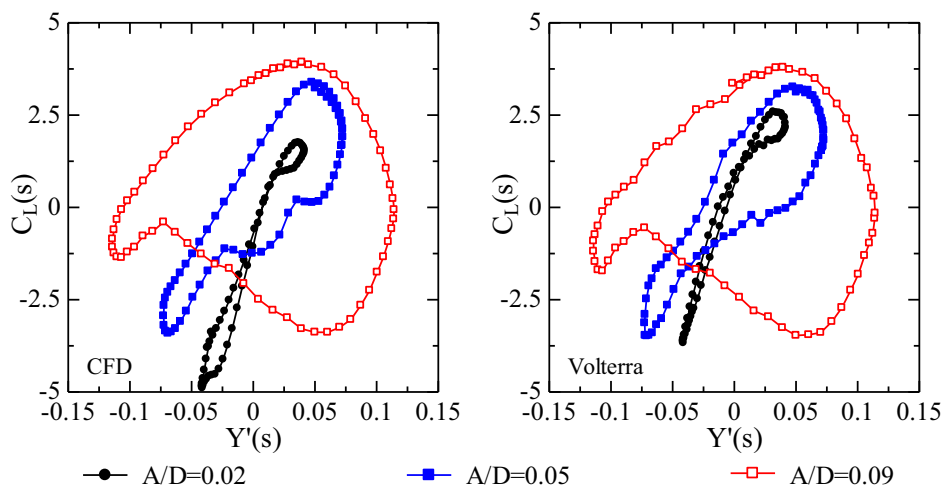
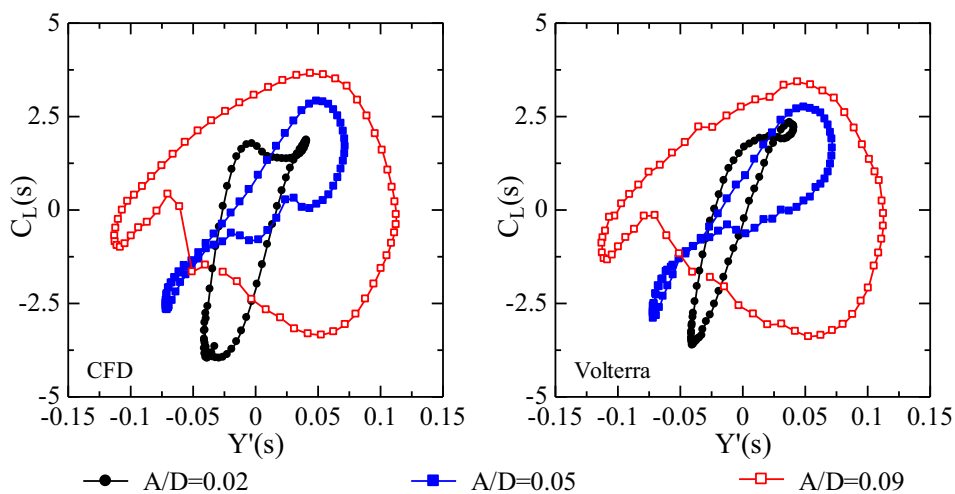
(a) Reduced velocity: $U/(\omega_n D) = 1.07$ (b) Reduced velocity: $U/(\omega_n D) = 1.22$ (c) Reduced velocity: $U/(\omega_n D) = 1.24$

Fig. 8. Comparison between sparse Volterra modeling results and CFD results. (a) Reduced velocity: $U/(\omega_n D)=1.07$. (b) Reduced velocity: $U/(\omega_n D)=1.22$. (c) Reduced velocity: $U/(\omega_n D)=1.24$.

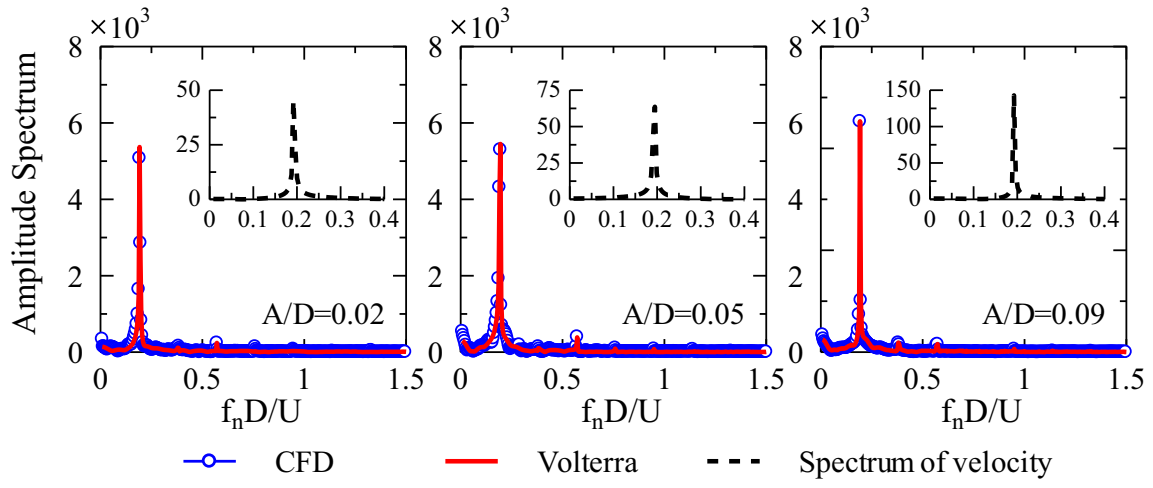
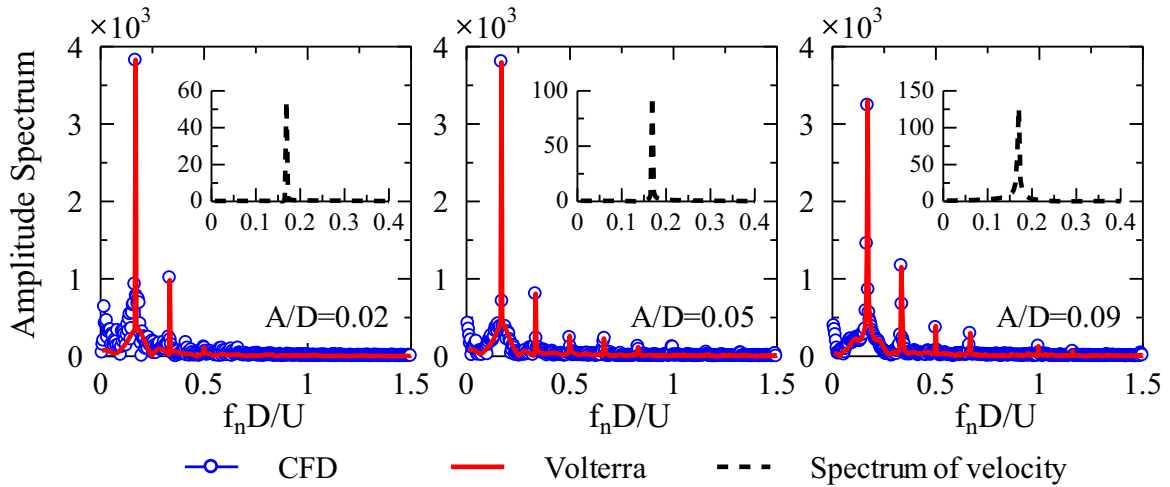
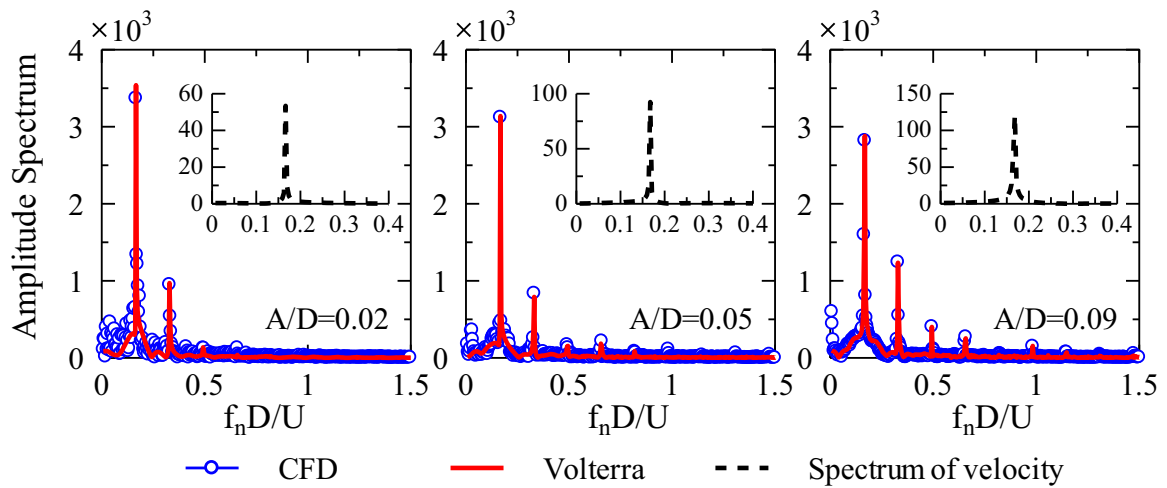
(a) Reduced velocity: $U/(\omega_n D) = 1.07$ (b) Reduced velocity: $U/(\omega_n D) = 1.22$ (c) Reduced velocity: $U/(\omega_n D) = 1.24$

Fig. 9. Comparison between sparse Volterra modeling results and CFD results. (a) Reduced velocity: $U/(\omega_n D)=1.07$. (b) Reduced velocity: $U/(\omega_n D)=1.22$. (c) Reduced velocity: $U/(\omega_n D)=1.24$.

Table 1
Mass-damping conditions adopted in the calculation.

Case	Mode number	Equivalent modal mass (kg/m)	Frequency (Hz)	Damping ratio (%)
A	1st vertical	48,023	0.079	0.2
B				0.5
C	10th vertical	26,393	0.324	0.2
D				0.5

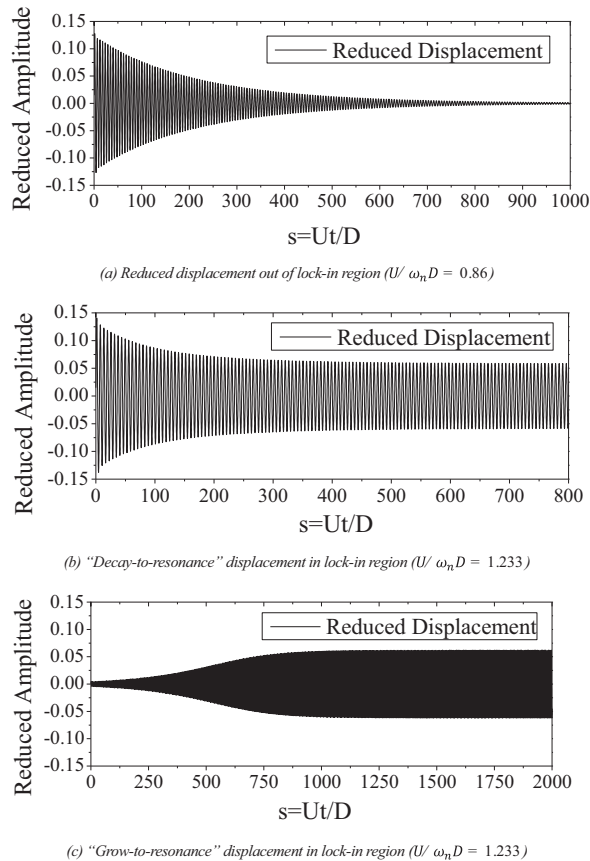


Fig. 10. Simulated VIV responses by sparse Volterra model. (a) Reduced displacement out of lock-in region ($U/\omega_n D=0.86$). (b) "Decay-to-resonance" displacement in lock-in region ($U/\omega_n D=1.233$). (c) "Grow-to-resonance" displacement in lock-in region ($U/\omega_n D=1.233$).

Volterra model. This may be due to the difference between the initial conditions of these two approaches. In the Volterra model approach, the initial condition is set to a small value of structural velocity and the VIV gradually develops through the accumulation of the "self-excited" effects. While, in the CFD approach, the VIV is activated by the "forcing effects" due to the shedding vortices in a fully developed flow field. In fact, the Volterra approach cannot simulate the excitation effects generated by shedding vortices when the structure is still, even though its contribution to the total VIV force can be ignored when the VIV is fully developed. This is also reflected in Fig. 7 that the simulating performance of the Volterra model at large-amplitude conditions ($A/D=0.05$ and $A/D=0.09$) is better than that at small-amplitude conditions ($A/D=0.02$). Since the initial conditions of these two approaches are different, the lengths of the "transient" stage could be varied to some extent.

When assessing the VIV performance of a bluff body, we mainly focus on its steady-state vibrations. Once the VIV is fully developed, the simulated results by the Volterra model can compare well with those of CFD approach.

The steady-state VIV amplitudes at four mass-damping cases calculated by the Volterra model are compared with the CFD results at the same mass-damping condition, as illustrated by Fig. 12. The results indicate that the VIV responses vary with the mass-damping parameter, and both the VIV amplitudes and the range of lock-in region decrease with the increase of mass-damping parameter. The calculated VIV amplitudes by Volterra model are comparable with the CFD results at all these four mass-damping conditions. This indicates that the Volterra model with kernels identified through forced-vibration data can be used to predicate the free-vibration responses of VIV at different mass-damping conditions, which is beyond the capacity of the traditional semi-empirical models.

However, the calculated range of lock-in region by the Volterra model is a little bit larger than that by the CFD approach. This may be due to the fact that in the last part of the lock-in region (wind velocities that are higher than the most unfavorable wind velocity) the VIV is sensitive to the phase lag between the high-frequency and the fundamental frequency components, which changes with the increase of vibration amplitude. At some certain wind velocities in the last part of the lock-in region, the phase lag versus structural status is quite complex that the VIV cannot fully develop and the structure exhibits "beat" phenomenon. The subtle relationships between the phase lag and the structural status at these certain velocities cannot be well reproduced by the Volterra model, since the kernels are identified through data at discrete vibration amplitudes. Thus, the identified kernels are in fact a comprehensive result with information from all force-vibration conditions due to the least-squares concept.

Taking Case B as an illustration, the calculated result at wind velocity in the last part of the lock-in region by the Volterra model is limit cycle oscillation, while the result by CFD approach is the "beat" phenomenon, as illustrated in Fig. 13. The insufficient modeling performance of the Volterra model at some wind velocities in the last part of the lock-in region can be improved once the kernels contain enough information at these conditions.

4. Calculating the VIV of Xihoumen Bridge

4.1. Technical roadmap for bridge VIV calculation

As illustrated in the above section, the Volterra approach can be used to simulate the free-vibration performance of VIV with kernels identified through forced-vibration data. This indicates that the Volterra model has captured the dominating inherent nature of fluid-structure interactions by using these identified kernels. In this section, the Volterra model is combined with the structural finite element method for calculating the VIV performance of the real bridge.

As mentioned above, the VIV response of a sectional model varies with that of a real bridge, since a real bridge is a flexible structure with 3D modal shapes, while a sectional model is a rigid structure. To take the influence of modal shapes into account, the Volterra model can be utilized to model the VIV forces on each segment of the deck along the span-wise, and the simulated forces can then be applied on the structural finite element model along the span-wise direction to calculate the structure's motion. Firstly, the initial condition should be given, which is generally selected to be a small value of structural amplitude or velocity, which is spatially distributed along the span-wise direction following the modal shape of a selected mode. Then, the VIV forces at each span-wise segment of the bridge are generated through the ROM approach with the given initial condition, and are applied on the structural finite element model to calculate the first step of structural motions. The calculated structural motions are then fed back to the ROM approach to generate the next step of VIV forces on each segment of the structure along the span-wise direction. By iterating the above steps, the bridge's response can be generated step by step, and the VIV performance of a bridge can be obtained after calculating the VIV response at different wind velocities for each mode.

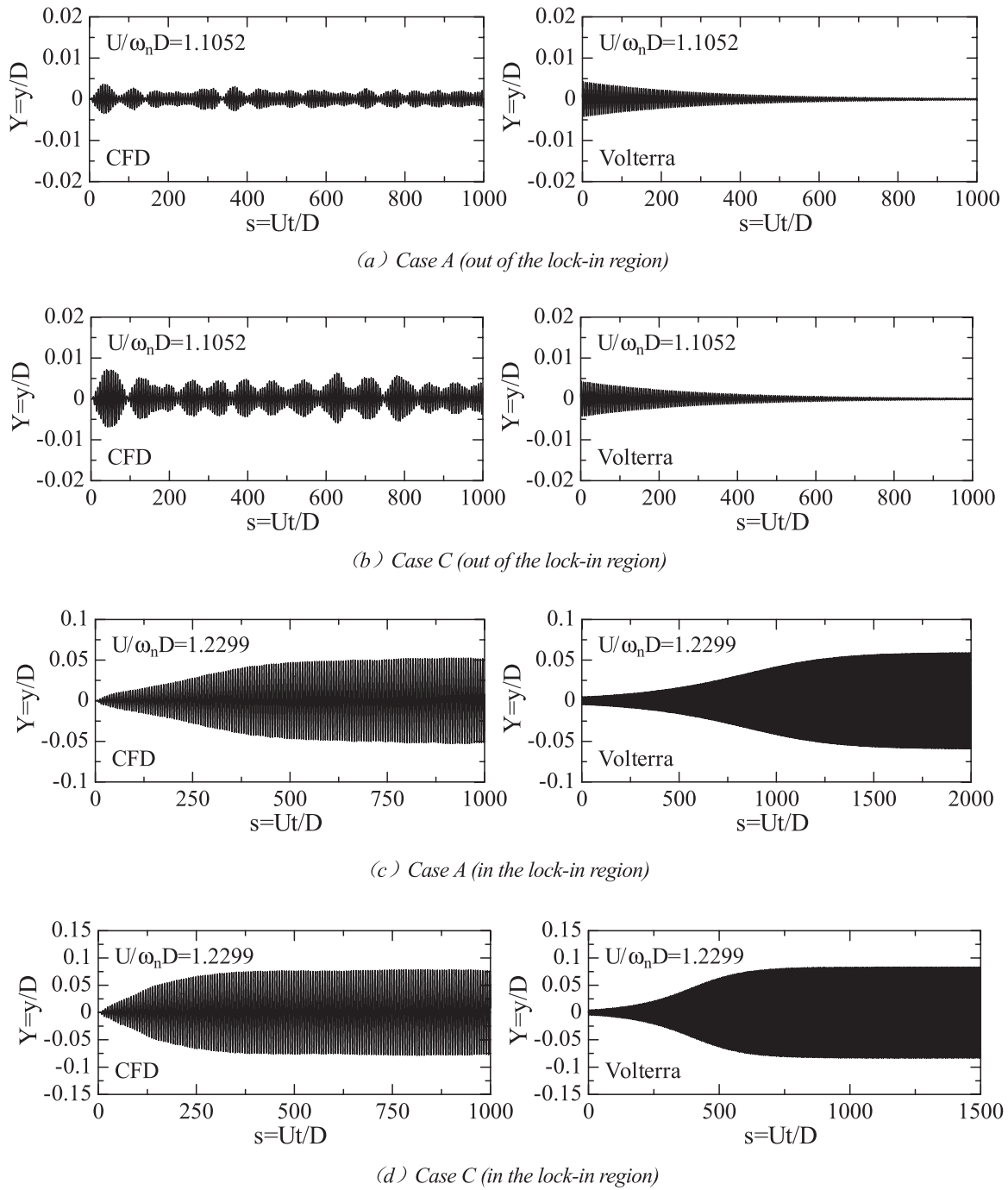


Fig. 11. Free-vibration displacements generated by ROM approach and CFD approach. (a) Case A (out of the lock-in region). (b) Case C (out of the lock-in region). (c) Case A (in the lock-in region). (d) Case C (in the lock-in region).

The calculation of structural motion at each iteration step can be fulfilled in the physical coordinate system, where the span-wise VIV forces are directly applied on the structural finite element model, or conducted in the modal coordinate system, since the bridge's VIV often occurs in a single mode. When conducting the calculation in modal coordinates, the modal generalized force should be generated at each time step. The technical roadmap for calculating the bridge's VIV is illustrated by Fig. 14.

4.2. Example and validation

To illustrate and validate the above established procedure, the Xihoumen Bridge is adopted here as a case study. The Xihoumen

Bridge experienced single mode VIV in several vertical bending modes at moderate wind speeds in the range of 5–13 m/s after erection but before being opened to the public. Thus, the vertical bending modes of this bridge are first generated using the finite element method. The finite element model is shown in Fig. 15.

The first 50 modes of the Xihoumen Bridge are calculated, among which there are 19 vertical bending modes. The modal shapes of the vertical bending modes, which have experienced VIV, according to field measurements (Li et al., 2014), are illustrated in Fig. 16.

The vertical bending mode VIVs of the Xihoumen Bridge are calculated using the approach established in Section 4.1. The calculation of the equation of motion is conducted in modal coordinate, thus the transition between the physical coordinate and the modal coordi-

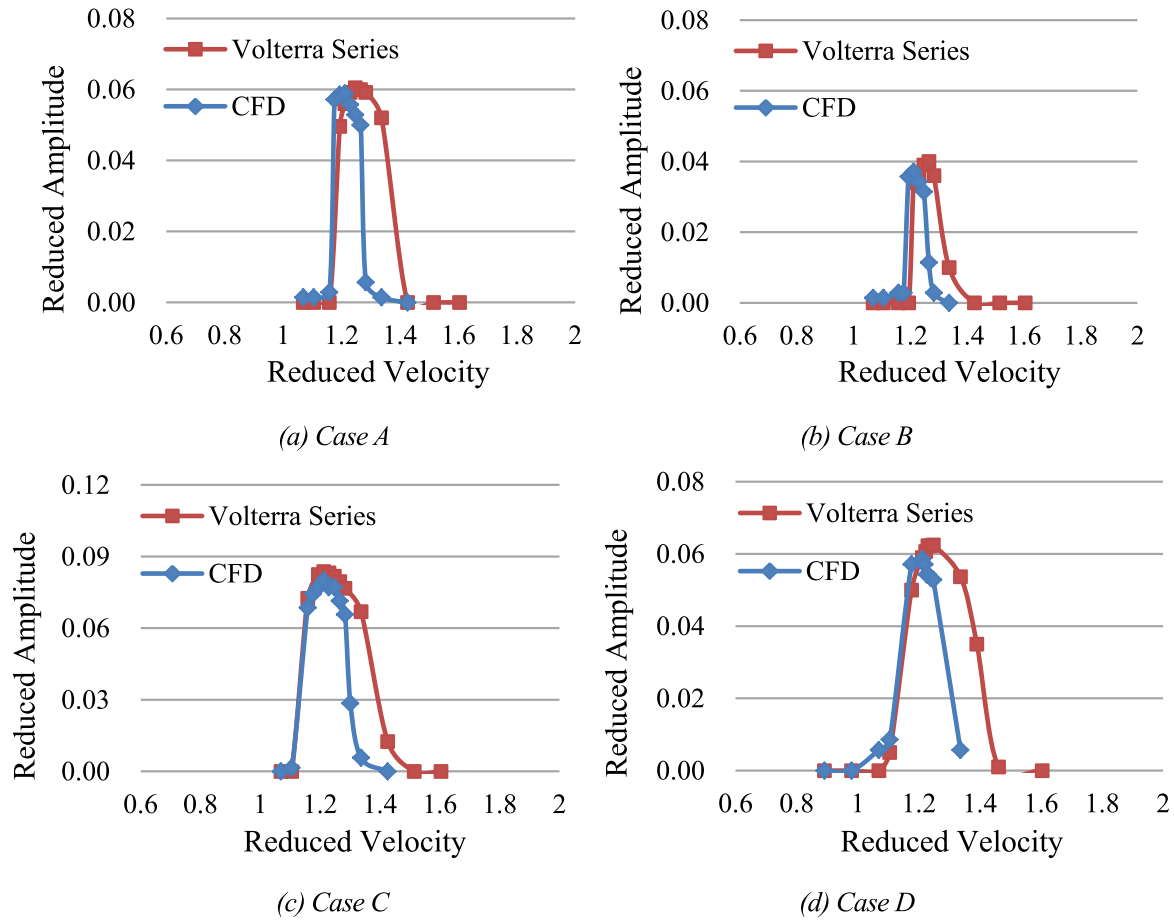


Fig. 12. Steady-state VIV amplitudes generated by Volterra approach and CFD approach. (a) Case A (b) Case B. (c) Case C (d) Case D.

nate is needed at each time step. The initial condition is set to $[\eta(0)=0, \eta'(0)=0.005]^T$, where $\eta(s)$ denotes the generalized coordinate of the selected mode. The time interval of the calculation is 0.1 s, and the calculation time length is 4000 s to make the VIV fully developed.

The steady-state VIV amplitudes of the first 20 vertical bending modes are obtained through the above procedure. Fig. 17 illustrates the comparison between the calculated results and those of field measurement. The calculated VIV amplitudes are qualitatively comparable with those of field measurement. However, the calculated velocity range of lock-in is a little bit smaller than that of field measurement for each mode. This may be due to the inconsistency between the CFD approach adopted in this study for kernel identification and the actual condition. Since the deck shape simulated in the CFD approach does not have railings while the actual deck does. Moreover, the calculation domain in the CFD approach is two-dimensional, which cannot consider the three-dimensionality of the flow fields and the influences of discretely distributed cross beams used for connecting the twin boxes.

Nevertheless, since the purpose of this study is to illustrate and validate the established approach for calculating the VIV of bridges by

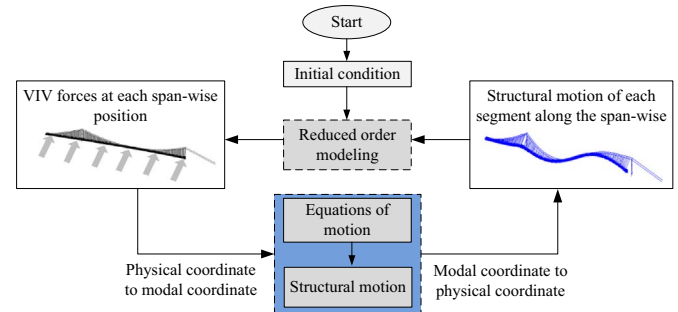


Fig. 14. Technical roadmap for bridge VIV calculation.



Fig. 15. Finite element model of Xihoumen Bridge.

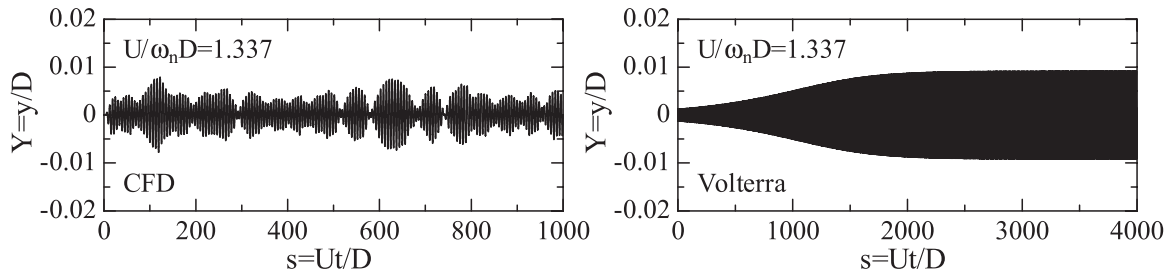


Fig. 13. Inconsistency between Volterra approach and CFD approach at certain wind velocity (Case B).

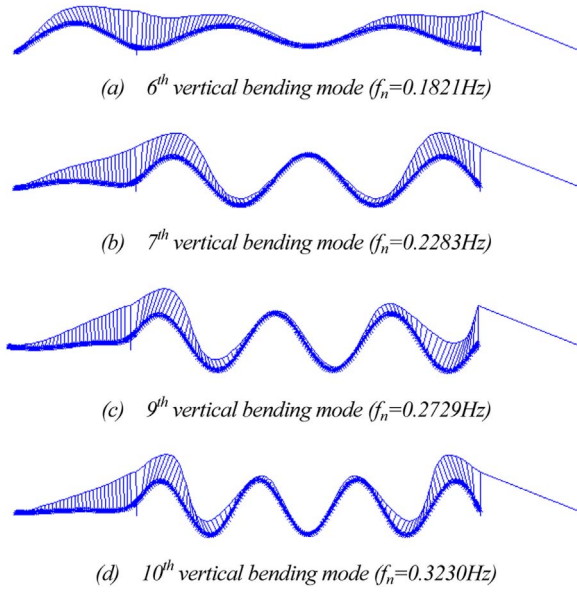


Fig. 16. Modal shapes of vertical bending modes that have experienced VIV. (a) 6th vertical bending mode ($f_n=0.1821$ Hz). (b) 7th vertical bending mode ($f_n=0.2283$ Hz). (c) 9th vertical bending mode ($f_n=0.2729$ Hz). (d) 10th vertical bending mode ($f_n=0.3230$ Hz).

using ROM technique, the identified kernels through data generated by 2D CFD calculations are adopted. In the future study, the kernels in Volterra model can be identified through data generated by wind tunnel experiments on sectional models or 3D CFD calculations by considering the influences of railings and distributed cross beams. In this case, the accuracy of the ROM approach can be expected to be obviously improved.

It should be noted that the span-wise coherence of VIV forces is not considered in this study. The involving of spatial coherence of VIV forces will reduce the VIV amplitude or even the range of lock-in region. Thus, the consideration of span-wise coherence of VIV forces should be based on an accurate expression for the spatial coherence function. Once the spatial coherence function of VIV forces is obtained by wind tunnel experiments, its influence can be taken into account in this approach by introducing a “correcting function” in the process of generating the modal generalized force at each time step, as done by Ehsan and Scanlan (1990).

Appendix A

Explicit expressions for the input matrix $\mathbf{Y}'_{N \times Q}$ in Eq. (6):

$$\mathbf{Y}'_{N \times Q} = \left[\mathbf{Y}'^1_{N \times Q_1}, \mathbf{Y}'^2_{N \times Q_2}, \dots, \mathbf{Y}'^p_{N \times Q_p}, \dots, \mathbf{Y}'^M_{N \times Q_M} \right] \quad (\text{A-1})$$

where the first term:

$$\mathbf{Y}'^1_{N \times Q_1} = \begin{bmatrix} Y'[0] & 0 & \dots & 0 \\ Y'[1] & Y'[0] & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ Y'[n] & Y'[n-1] & \dots & Y'[n-M_1] \end{bmatrix} \quad (\text{A-2})$$

$Q_1 = M_1 + 1$

the second term:

$$\mathbf{Y}'^2_{N \times Q_2} = \begin{bmatrix} Y'[0]^2 & 0 & \dots & 0 & \dots & 0 & 0 & \dots & 0 \\ Y'[1]^2 & Y'[1]Y'[0] & \dots & 0 & \dots & Y'[1-M_2]Y'[1] & Y'[1-M_2]Y'[0] & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ Y'[n]^2 & Y'[n]Y'[n-1] & \dots & Y'[n]Y'[n-M_2] & \dots & Y'[n-M_2]Y'[n] & Y'[n-M_2]Y'[n-1] & \dots & Y'[n-M_2]^2 \end{bmatrix} \quad (\text{A-3})$$

$Q_2 = (M_2 + 1)^2$

the pth term:

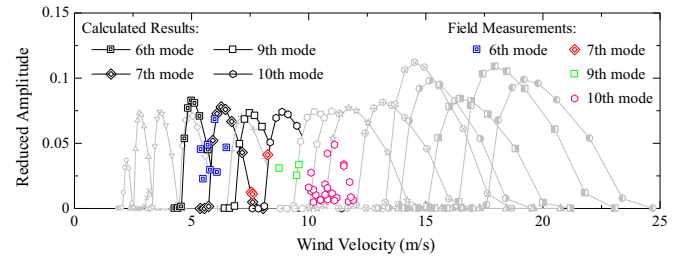


Fig. 17. Calculated VIV performances of Xihoumen Bridge and comparison with field measurements.

5. Conclusions

The development of reduced order modeling techniques offers a possibility to model the complex fluid-structure interactions existing in the VIV of bridge decks, which is beyond the capacity of traditional semi-empirical models. In this paper the performance of Volterra theory-based reduced-order modeling in simulating the VIV of bridge decks is examined and validated by comparison with a CFD approach involving a real deck shape. Results indicate that the Volterra approach can well simulate the dominant inherent nature of fluid-structure interactions existing in the VIV of bridge decks. An approach for calculating the VIV of bridges is also proposed. The VIV performance of a real bridge is obtained by using this approach and is compared with field measurements. Spatial coherence of VIV forces is not considered in this study. However, once the spatial coherence function of VIV forces is obtained by wind tunnel experiments, its influence can be easily taken into account by introducing a “correcting function” to this existing approach.

Acknowledgements

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$$Y'_{N \times Q_p} = \begin{bmatrix} Y'[0]^p & 0 & \cdots & 0 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ Y'[1]^p & Y'[1]^{p-1}Y'[0] & \cdots & Y'[1]^{p-1}Y'[1-M_p] & \cdots & Y'[1-M_p]^{p-1}Y'[1] & Y'[1-M_p]^{p-1}Y'[0] & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ Y'[n]^p & Y'[n]^{p-1}Y'[n-1] & \cdots & Y'[n]^{p-1}Y'[n-M_p] & \cdots & Y'[n-M_p]^{p-1}Y'[n] & Y'[n-M_p]^{p-1}Y'[n-1] & \cdots & Y'[n-M_2]^p \end{bmatrix}$$

$Q_p = (M_p+1)^p$

(A-4)

Explicit expressions for the kernel vector $\mathbf{H}_{Q \times 1}$ in Eq. (6):

$$\mathbf{H}_{Q \times 1} = \left[\mathbf{H}_{Q_1}^1, \dots, \mathbf{H}_{Q_p}^p, \dots, \mathbf{H}_{Q_M}^M \right]^T$$
(A-5)

where the first term:

$$\mathbf{H}_{Q_1}^1 = \underbrace{[h[0], h[1], \dots, h[M_1]]}_{Q_1 = M_1 + 1}$$
(A-6)

the second term:

$$\mathbf{H}_{Q_2}^2 = \underbrace{[h[0, 0], h[0, 1], \dots, h[0, M_2], \dots, h[M_2, 0], h[M_2, 1], \dots, h[M_2, M_2]]}_{Q_2 = (M_2+1)^2}$$
(A-7)

the pth term:

$$\mathbf{H}_{Q_p}^p = \underbrace{[h[0, 0, \dots, 0], h[0, 0, \dots, 1], \dots, h[0, 0, \dots, M_p], \dots, h[M_p, M_p, \dots, 0], h[M_p, M_p, \dots, 1], \dots, h[M_p, M_p, \dots, M_p]]}_{Q_p = (M_p+1)^p}$$
(A-8)

Explicit expressions for the input matrix $\mathbf{Y}'_{N \times Q}$ in Eq. (9):

where the first term:

$$Y'_{N \times Q_1} = \underbrace{\begin{bmatrix} Y'[0] & 0 & \cdots & 0 \\ Y'[1] & Y'[0] & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ Y'[n] & Y'[n-1] & \cdots & Y'[n-M_1] \end{bmatrix}}_{Q_1 = M_1 + 1}$$
(A-9)

the second term:

$$Y'_{N \times Q_2} = \underbrace{\begin{bmatrix} Y'[0]^2 & 0 & \cdots & 0 \\ Y'[1]^2 & Y'[0]^2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ Y'[n]^2 & Y'[n-1]^2 & \cdots & Y'[n-M_2]^2 \end{bmatrix}}_{Q_2 = M_2 + 1}$$
(A-10)

the pth term:

$$Y'_{N \times Q_p} = \underbrace{\begin{bmatrix} Y'[0]^p & 0 & \cdots & 0 \\ Y'[1]^p & Y'[0]^p & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ Y'[n]^p & Y'[n-1]^p & \cdots & Y'[n-M_p]^p \end{bmatrix}}_{Q_p = M_p + 1}$$
(A-11)

Explicit expressions for the kernel vector $\mathbf{D}_{Q \times 1}$ in Eq. (9):

$$\mathbf{D}_{Q \times 1} = \left[\mathbf{D}_{Q_1}^1, \dots, \mathbf{D}_{Q_p}^p, \dots, \mathbf{D}_{Q_M}^M \right]^T$$
(A-12)

where the first term:

$$\mathbf{D}_{Q_1}^1 = \underbrace{[h[0], h[1], \dots, h[M_1]]}_{Q_1 = M_1 + 1}$$
(A-13)

the second term:

$$\mathbf{D}_{Q_2}^2 = \underbrace{[h[0, 0], h[1, 1], \dots, h[M_2, M_2]]}_{Q_2 = M_2 + 1}$$
(A-14)

the pth term:

$$\mathbf{D}_{Q_p}^p = \underbrace{[h[0, \dots, 0], h[1, \dots, 1], \dots, h[M_p, \dots, M_p]]}_{Q_p = M_p + 1}$$
(A-15)

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